



NFL - Contract Value and Offseason Evaluation Methods

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NFL Contract Value and Offseason Evaluation Methods – Context

- **Context:** After each season, NFL front offices and executives are faced with tough decisions in whether they should extend, retain, release, or re-evaluate certain players based off their contracts (in regards to the salary cap) and statistics. The purpose of this project was to provide NFL organizations with tools to help them quantify and evaluate these players. By calculating a value gap between a player's predicted Adjusted Average Salary per Year and their actual Adjusted Average Salary per Year, front offices could identify which players were outperforming and underperforming their current contracts. Additionally, creating a model using surplus of expected value (total EPA – Expected EPA) would help front offices and executives group players into categories of whether they should extend, retain, release, or re-evaluate players. Using these two models above, NFL executives would have plenty of tools at their disposal to help with offseason decision-making.
- **Dataset:** The two datasets used during this process were from OvertheCap and NFLFastR. The datasets were joined together and only the 2018-2024 seasons were included in the models.
- **Machine Learning Models Used:** Linear Regression and Gradient Boosting Regressions were the two models used during this process. Linear Regression worked very well predicting Adjusted Average Salary per Year, while Gradient Boosting provided a good way to handle non-linear relationships, capture feature interactions, and handle skewed metrics for predicting surplus of expected value (total EPA – Expected EPA). Normalizations techniques were used to ensure that the most important features used were valued at the same scale. R-Squared was used to evaluate the models as well.
- **Future Implications:**
 - How can projected surplus value help optimize extension and release value?
 - What are some warning signs or criteria to look for when a team should avoid signing a player to an extension?
 - How do these models influence draft strategy?
 - How would this model help identify the metrics that forecast early signs of decline or likelihood of a contract extension?

NFL Contract Value and Offseason Evaluation Methods – Input Variables

- **Input Variables:** The variables used in this model include a combination of efficiency, production, and career context. Again, the purpose of using these variables was to evaluate player performance and cost-effective practices for the future. Some of the most important metrics, are noted below:

total_epa	Total Expected Points Added throughout the season
avg_epa_per_play	Average Expected Points Added per Play
cpoe_avg	Average Completion Percentage Over Expected
epa_per_million	Quantifies the total EPA produced per one million dollars
tds_per_million	Quantifies the number of TD's produced per one million dollars
td_per_int	Touchdowns / Interceptions
rush_success_rate	Percentage of rushing players that are deemed successful
yac_epa_total	Total Expected Points Added after a catch
apy_cap_pct	Average Per Year as a percentage of the team's salary cap. It quantifies the amount a player's salary takes up as a percentage of the team's total salary cap.

NFL Contract Value and Offseason Evaluation Methods – Target and Created Variables

- **Target and Created Variables:** The target variables used in the two models during this project, along with the most important columns created in the feature engineering phase, are noted below:

inflated_apv	Average Salary Per Year after adjusting for changes in the salary cap and being scaled to a common salary cap environment
predicted_apv	Model generated prediction of what a player should make per season
value_gap	Predicted APY – Inflated APY. How much more or less a player should be paid
expected_epa	Expected total EPA for a season based off their current contract and statistics
surplus_value	Total EPA – Expected total EPA. How much more or less a player produced during a season compared to what was expected.
epa_z	Normalized Expected Points Added per Million Dollars Earned
yards_z	Normalized Total Yards per Million Dollars Earned
tds_z	Normalized Total Touchdowns per Million Dollars Earned

NFL Contract Value – Model Output (QB – Linear Regression)

- **Model Context:** A Linear Regression model was used to calculate predicted APY for quarterbacks from the 2018-2024 season. Features used in the model included "total_epa", "total_yards", "epa_z", "tds_z", "td_per_int", "qb_epa_total", "avg_epa_per_play", "cpoe_avg", "rush_epa_total". After using these features in the model to predict APY, a value gap metric was created to represent the difference between predicted APY and actual APY. With this metric, the five most underpaid and overpaid quarterbacks were identified and are noted below.

- **Top 5 Most Underpaid QBs:**

Player	Season	Inflated APY (millions)	Predicted APY (millions)	Value Gap
J. Love	2023	\$16.76	\$46.44	\$29.68
C. Stroud	2023	\$11.26	\$40.84	\$29.58
J. Herbert	2020	\$9.36	\$37.59	\$28.23
B. Mayfield	2023	\$4.96	\$30.33	\$25.37
T. Brady	2020	\$35.21	\$59.97	\$24.76

- **Top 5 Most Overpaid QBs:**

Player	Season	Inflated APY (millions)	Predicted APY (millions)	Value Gap
D. Watson	2022	\$61.69	\$16.56	(\$45.13)
D. Prescott	2024	\$65.59	\$20.61	(\$44.98)
J. Burrow	2023	\$68.31	\$26.85	(\$41.46)
T. Lawrence	2024	\$60.13	\$21.15	(\$38.98)
D. Jones	2023	\$49.68	\$13.52	(\$36.16)

NFL Contract Value – Model Output (RB – Linear Regression)

- **Model Context:** A Linear Regression model was used to calculate predicted APY for running backs from the 2018-2024 season. Features used in the model included "total_epa", "total_yards", "epa_z", "tds_z", "yds_carry", "total_tds", "rush_epa_total", "rush_success_rate", "avg_epa_per_play", "rec_yards", and "yac_epa_total". After using these features in the model to predict APY, a value gap metric was created to represent the difference between predicted APY and actual APY. With this metric, the five most underpaid and overpaid running backs were identified and are noted below.

- **Top 5 Most Underpaid RBs:**

Player	Season	Inflated APY (millions)	Predicted APY (millions)	Value Gap
D. Henry	2024	\$8.74	\$14.81	\$6.07
J. Gibbs	2023	\$5.54	\$10.72	\$5.18
N. Harris	2021	\$4.99	\$9.72	\$4.73
C. Hyde	2019	\$4.15	\$8.59	\$4.44
S. Barkley	2024	\$13.76	\$17.92	\$4.16

- **Top 5 Most Overpaid RBs:**

Player	Season	Inflated APY (millions)	Predicted APY (millions)	Value Gap
L. Bell	2019	\$19.47	\$10.24	(\$9.23)
E. Elliott	2019	\$22.25	\$14.20	(\$8.05)
N. Chubb	2021	\$18.66	\$11.31	(\$7.35)
D. Johnson	2018	\$20.48	\$13.81	(\$6.67)
T. Gurley	2018	\$22.64	\$16.08	(\$6.56)

NFL Evaluation Method – Model Output (QB – Gradient Boosting)

- **Model Context:** A Gradient Boosting Regression model was used to calculate predicted surplus value for quarterbacks from the 2018-2024 season. Features used in the model included "total_yards", "avg_epa_per_play", "epa_z", "tds_z", "inflated_apy", "apy_cap_pct", "draft_round", "draft_overall", "td_per_int", "qb_epa_total", "cpoe_avg", and "rush_epa_total". Surplus value was a metric created in feature engineering based off subtracting total EPA and Expected EPA. Using this metric, the player is grouped into four categories: Extend (top 10%), Retain (top 40%), Evaluate (top 75%), and Replace/Release (Bottom 25%).

- **Top 5 Extend Candidates QBs:**

Player	Season	Predicted Surplus	Offseason Move
J. Allen	2021	178.89	Extend
P. Mahomes	2020	178.38	Extend
T. Brady	2020	176.34	Extend
D. Brees	2018	171.66	Extend
R. Tannehill	2020	169.43	Extend

- **Top 5 Release Candidates QBs:**

Player	Season	Predicted Surplus	Offseason Move
E. Stick	2023	-20.78	Replace/Release
D. Jones	2024	-22.12	Replace/Release
A. O'Connell	2023	-23.73	Replace/Release
D. Watson	2022	-24.63	Replace/Release
J. Flacco	2022	-24.71	Replace/Release

NFL Evaluation Method – Model Output (RB – Gradient Boosting)

- **Model Context:** A Gradient Boosting Regression model was used to calculate predicted surplus value for running backs from the 2018-2024 season. Features used in the model included "inflated_apy", "total_epa", "avg_epa_per_play", "epa_z", "tds_z", "yac_total", "rush_attempts", "rush_yards", "rush_tds", "rush_epa_total", "rush_success_rate", "rush_epa_total", "avg_rush_epa", "rec_yards", "yac_epa_total", and "fumbles". Surplus value was a metric created in feature engineering based off subtracting total EPA and Expected EPA. Using this metric, the player is grouped into four categories: Extend (top 10%), Retain (top 40%), Evaluate (top 75%), and Replace/Release (Bottom 25%).

- **Top 5 Extend Candidates RBs:**

Player	Season	Predicted Surplus	Offseason Move
S. Barkley	2024	27.72	Extend
D. Henry	2024	26.82	Extend
D. Henry	2020	25.81	Extend
T. Gurley	2018	19.19	Extend
B. Irving	2024	13.65	Extend

- **Top 5 Release Candidates RBs:**

Player	Season	Predicted Surplus	Offseason Move
J. Jacobs	2024	-30.12	Replace/Release
T. Pollard	2024	-31.88	Replace/Release
A. Peterson	2018	-31.95	Replace/Release
A. Peterson	2019	-32.01	Replace/Release
E. Mitchell	2021	-32.33	Replace/Release

NFL Contract Value and Offseason Evaluation Methods – Conclusion

Summary:

- Advanced metrics were used by a Linear Regression model to predict Adjusted Average Salary per Year and identify which players (QBs and RBs) were outperforming and underperforming their current contracts.
- Similarly, a Gradient Boosting model provided a good way of predicting surplus value (total EPA – Expected EPA), which would help front offices and executives group players into categories of whether they should extend, retain, release, or re-evaluate players.
- By predicting NFL contract value and surplus of expected value, NFL executives would have plenty of tools at their disposal to help support their offseason decision-making.

Key Takeaways:

- EPA is a Strong Driver to Contract Direction: Quarterbacks with strong and consistent EPA metrics generally were predicted into the “Extend” category. Similarly, running backs with high EPA metrics in rushing and receiving, along with high rushing success rates, were commonly projected into the “Extend” category.
- Pure Rush EPA is Not the Only Determining Factor for RB Performance: As previously stated, not only was rushing EPA an important metric, but receiving EPA for RB’s was also a very valuable statistic for determining contract value and performance in the future.
- Salary Cap Efficiency is as Important as Raw EPA: Using metrics such as EPA Per \$Millions and Touchdown per \$Millions proved to be extremely crucial for determining extend and release candidate players. For instance, a player could have an average EPA value but expensive contract, making the player a potential release prospect.

Next Steps:

- Please keep in mind that this project only examined the statistics of quarterbacks and running backs. No multi-year carry over or trend information were used to analyze statistics and certain contract situations. In the future, this is something I am hoping to incorporate into the models.
- Identify what statistics are the most important year over year to continue assessing player value and improving the model in the future.